

AUTONOMOUS FINTECH ASSISTANT

Monevo: Voice-First WhatsApp AI Financial Agent

"Talk to your ledger like you talk to a friend."

A zero-friction, voice-powered financial copilot built inside WhatsApp. Turning natural speech, colloquial accents, and localized slang into real-time double-entry ledgers, dynamic digital receipts, and visual analytics.

SPEECH INTELLIGENCE
AssemblyAI Universal-3.5-pro

INTERFACE
Meta WhatsApp Cloud API

EXECUTION ENGINE
Node.js • TypeScript • MongoDB

Bookkeeping is Broken for Fast-Paced Commerce

Over 300 million small merchants, freelancers, and gig workers operate completely on the move. Traditional bookkeeping software fails them.



High Cognitive Overhead

Traditional accounting apps demand opening dashboards, remembering passwords, navigating multi-level menus, and filling out 6+ form fields per expense. Users abandon them in days.

Complex Forms

Dashboard Fatigue



Deferred & Lost Transactions

Transactions happen during traffic, vendor deliveries, or market haggling. Because typing is inconvenient on the move, 70%+ of transactions are postponed and permanently forgotten.

Unrecorded Cashflow

Profit Leakage



Accents & Regional Slang Barriers

Standard speech engines choke on regional dialects, colloquial currency names ("5k", "kobo", "Naira"), and local expenses ("light bill", "POS charge", "fuel"). Voice assistants couldn't understand them—until now.

Dialect Failure

Context Blindness

Monevo: Zero App. Zero Login. Just Speak on WhatsApp.

Monevo converts the universal WhatsApp voice note into a full-featured financial ERP copilot.



1. Natural Voice Input

Hold the WhatsApp mic and talk naturally: *"Spent 15k on fuel at Total"* or *"Client David paid 200k for brand design"*. No rigid syntax required.

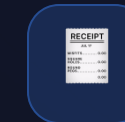
✓ Hands-free while driving or working



2. AssemblyAI Audio Brain

Audio stream is instantly processed with AssemblyAI flagship models using contextual prompt injection and customized fintech word-boost dictionaries.

✓ Accents, noise & slang parsed accurately



3. Instant Multimodal Delivery

Returns immediate transaction confirmation, voice synthesis response, on-the-fly downloadable branded SVG/PNG receipts, and visual spending analytics.

✓ Receipts & charts delivered in chat

Supercharged by AssemblyAI Universal Speech API

How Monevo utilizes AssemblyAI's advanced speech understanding parameters to achieve near-flawless transcription on noisy real-world WhatsApp audio.

🔧 Production Configuration in `src/services/assemblyai/`

```
// AssemblyAI Universal Flagship Transcribe
await client.transcripts.transcribe({
  audio: audioBuffer,
  speech_models: ['universal-3-5-pro', 'universal-2'],
  boost_param: 'high',
  word_boost: [
    'Monevo', 'Naira', 'kobo', 'POS', 'light bill',
    'spending analytics', 'receipt', 'fuel'
  ],
  prompt: 'WhatsApp voice note with financial transactions in Nigerian English, Naira
amounts, and receipt inquiries.',
  punctuate: true, format_text: true
});
```

🎯 Domain Word Boosting ('high')

Financial slang like "5k", "POS", and "kobo" are frequently misheard by vanilla speech engines. AssemblyAI's boost parameter forces high acoustic confidence for 40+ fintech terms.

🧠 Contextual Prompt Engineering

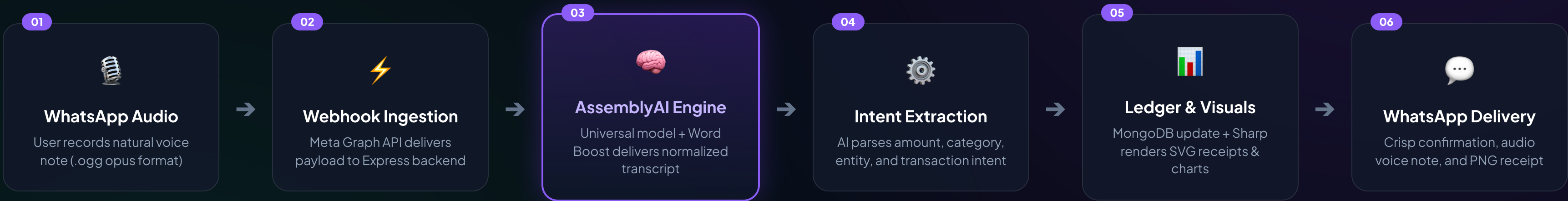
By injecting specialized financial context into AssemblyAI's prompt parameter, the speech engine primes acoustic decoders for spoken transaction narratives, vendor names, and rapid price statements.

🛡️ Production Resiliency & Key Sanitization

Built-in MIME validation (OGG/Opus WhatsApp format), 25MB payload guardrails, and automated API-key redaction wrappers to safeguard secrets across error boundaries.

End-to-End Conversational Pipeline

How voice packets travel from user speech to verified ledger updates and visual artifacts in under 2 seconds.



 **Autonomous Clarification Loop**
If audio misses required details (e.g. amount or vendor), Monevo pauses execution, prompts the user via voice/text, and remembers context.

stateful clarificationHandler

Four Core Superpowers in One WhatsApp Chat

Engineered for maximum utility with zero dashboard friction.



Voice-to-Ledger

Speaks: *"Spent 12k on fuel at Total"*
Instant categorization, deduction from balance, and MongoDB transaction entry.

```
recordTransaction()
```



Voice Clarification

User: *"Record my purchase for David"*
Monevo: *"How much did you pay David?"*
User answers via voice note and transaction finalizes.

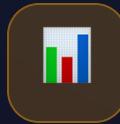
```
clarificationService()
```



Instant Branded Receipts

Speaks: *"Make receipt for Sarah 50,000 for consulting"*
Sharp engine renders an SVG receipt and delivers image to WhatsApp.

```
generateReceiptTool()
```



Visual Analytics

Speaks: *"What is my spending analysis for this week?"*
Delivers breakdown summary and graphical spending distribution charts.

```
analyticsService()
```

 Voice Note (0:04) — "Bought groceries 25k at Spar"

 **Expense Logged:** ₦25,000 (Groceries) • New Balance: ₦245,000 • *"Need a receipt for this?"*

Democratizing Financial Tracking for The Next Billion Users

WhatsApp has over 2.7 billion active users. Voice notes are the default communication medium in emerging economies.

2.7B+

Zero Onboarding Friction

Users already have WhatsApp open 20+ times a day. Monevo eliminates app store download drop-offs and eliminates device storage constraints on budget smartphones.

100%

Inclusive Voice Interface

Text keyboards and complex financial spreadsheets exclude millions of informal merchants. AssemblyAI's voice technology empowers anyone who can speak to keep audit-ready financial books.

5x

Higher Logging Retention

Because recording an expense takes under 4 seconds via a voice note (vs. 45 seconds on traditional apps), users maintain their ledger daily rather than abandoning it.

Built with Robust, Modern Engineering

100% TypeScript, enterprise test coverage, and modular microservices architecture.

🔧 Technology Stack

Speech-to-Text: AssemblyAI SDK

Backend: Node.js, Express, TS

Visuals: Sharp & SVG Engine

Messaging: Meta WhatsApp API

Database: MongoDB Atlas

Testing: Jest (Unit, E2E, Mocks)

🚀 What's Next on the Roadmap

- **AssemblyAI Streaming WebSockets:** Real-time live transcription feedback during recording.
- **WhatsApp Flows Integration:** Interactive in-chat forms and bank-sync connections.
- **Multi-Currency Autoconversion:** Cross-border ledger support (USD, EUR, GBP, NGN, KES).
- **Automated PDF Invoicing:** Voice-directed PDF invoice dispatch to clients.

Experience Monevo Today

Transforming voice into financial intelligence with AssemblyAI.

github.com/monevo-ai

READY FOR JUDGING